INCH-POUND
MIL-I-24092/10A
21 September 1993
SUPERSEDING
MIL-I-24092/10
4 May 1989

MILITARY SPECIFICATION SHEET

INSULATING VARNISH, WATER BASE, BAKING, FLEXIBLE, FOR DIP PROCESSING, GRADE WB, CLASS 130 TO 180 THERMAL CLASS

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24092.

REQUIREMENTS:

1. Grade - WB. Clear, water base baking varnish that cures to a flexible film.

2. Class - 130 to 180. The cured varnish shall meet the requirements of the specified thermal class when determined in accordance with MIL-I-24092.

3. Composition - III. Composition III varnishes are water reducible varnishes in which the solvent consists primarily of water along with lesser amounts of organic co-solvents. These are relatively complex solvent mixtures as compared to those used in the CB and CBH grade varnishes. Composition III varnishes offer higher flash points and reduced aromatic emission.

4. The varnishes shall meet the qualification and quality conformance requirements specified in tables I and II herein when tested in accordance with MIL-I-24092.

AMSC N/A FSC 5970
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TABLE I. Qualification requirements, grade WB

Characteristics (tests)	Units	Requirements
Storage life	Months, minimum	6
Nonvolatiles	Percent, minimum	35
Flash point	°C, minimum	90
Drying time at 150°C	Minutes, maximum	120
Viscosity: 20 r/min	Centipoise	200 to 1300
Build, as received	Mil, minimum	8.0
Dielectric strength:		
96/23/50	Volts per mil, minimum	2000
96/23/96	Minimum	1500
24/23/water	Minimum	1500
Bond strength		
at 23°C	Pounds force, minimum	20
at 150°C	Pounds force, minimum	1.0
Thermal endurance (M2, K2) only:		
Class 130	Temperature index, °C, minimum	130
Class 155	Temperature index, °C, minimum	155
Class 180	Temperature index, °C, minimum	180
Salt water proofness	Hours, minimum	100
Hydrolytic stability	Percent decrease in bond strength, maximum	To be determined
Chemical resistance:		
Hydraulic fluid	Percent decrease in bond strength, maximum	50
Lubrication oil		50
Cleaning fluid		50
Distilled water		50
Detergent solution		50

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TABLE I. Quality conformance requirements, grade W3

Characteristics (tests)	Units	Requirements
Nonvolatiles	Percent of qualification value	±50
Drying time at 150°C	Minutes, maximum	120
Viscosity at 20 r/min	Centipoise, percent of qualification value	±20
Build as received	Mil, minimum	0.9
Specific gravity	Percent of as-received value	±1.0
Bond strength at 23°C	Pounds force, minimum	20

NOTE:

Part or Identifying Number (PIN). Part number shall be M24092/10-(dash number indicating thermal class). For example, M24092/10-155 describes a material meeting the requirements of this specification and having class 155 thermal endurance. The PIN shall be the same format as specified in MIL-I-24092.

Custodians:
Army - ER
Navy - SH
Air Force - 11

Review activities:
Army - MI, AR
Navy - EC, YD, OS
Air Force - 99
DLA - GS, DS

User activities:
Army - ME
Navy - MC, AS

Preparing activity:
Navy - SH
(Project 5970-1082)

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